LESSON PLAN

LSF, STABILITY, A/C PERFORMANCE (6)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	31
Total No. of Lesson Plans	33	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Video on aircraft stability
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Video on aircraft stability
5. Specific Objectives	6.1.4.1 Trainee will be able to describe aircraft stability in all the three axis.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review video clips; Instructor supervises note-taking on stability axes; Safety: Ensure no loose items during video playback.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the three axes of an aircraft?
(b) Define static stability in your own words.
(c) Why is stability important for flight safety?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Longitudinal stability definition • Pitching moment characteristics • Center of gravity effects • Horizontal stabilizer function	L & Ds	PPT, IB	15 min
(b)	• Lateral stability definition • Rolling moment characteristics • Dihedral angle effect • Wing sweep influence	L & Ds	PPT, PDF	20 min
(c)	• Directional stability definition • Yawing moment characteristics • Vertical stabilizer function • Rudder control authority	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the horizontal stabilizer contributes to longitudinal stability.
(b) Describe the role of the dihedral angle in lateral stability.
(c) How does the vertical stabilizer affect directional stability?

12. Home Assignment and Follow Up	Prepare a brief note on how the concepts of aircraft stability discussed in this lesson will assist in understanding low-speed wind tunnel testing in LP 32.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Low speed wind tunnel and wind tunnel balances. Demonstration on pressure plotting experiments	Refer to the master syllabus reference list	LP 32

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min